

EnvFactory: Scaling Tool-Use Agents via Executable Environments Synthesis and Robust RL

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Abstract

Tool-use agents require diverse, well-specified execution environments to train effectively on realistic API interactions. Existing reinforcement learning frameworks treat environment definition as a manual, monolithic task, creating scalability bottlenecks for multi-tool agents. We introduce **EnvFactory**, a Python package that synthesizes executable task environments from API schemas, verifies environment correctness via stateful protocol validation, and generates calibrated trajectories for robust agent training. Our approach combines dependency inference, session-aware state management, and sandboxed execution to reduce environment authoring overhead by $\sim 60\%$ while maintaining semantic consistency across tool interactions. We demonstrate the framework on a synthetic multi-API benchmark and validate the integrity of generated trajectories against a reinforcement learning baseline, showing improved sample efficiency and task success rates compared to random exploration baselines.

1 Introduction

Autonomous agents that interact with external tools and APIs represent a frontier in AI systems research Brown et al. [2023], Schoop et al. [2023]. Training such agents requires two critical components: (1) a diverse set of executable environments that faithfully simulate API behavior, and (2) high-quality trajectories that guide exploration toward correct tool usage patterns. Current approaches suffer from fundamental limitations:

1. **Manual environment authoring** scales poorly. Each new API tool requires hand-crafted simulation code, domain-specific mocking logic, and stateful verification—tasks prone to inconsistency and maintenance burden.
2. **Trajectory quality is assumed, not validated.** Existing trajectory synthesis methods do not verify that generated sequences respect API constraints, preconditions, or data dependencies, leading to agent training on infeasible action sequences.
3. **No principled composition mechanism.** Multi-tool agents need environments where tool calls can be chained; existing frameworks lack first-class support for modeling inter-tool dependencies.

We address these gaps with **EnvFactory**, a framework that: (a) automatically synthesizes stateful execution environments from formal API specifications, (b) verifies trajectory feasibility via constraint checking and dependency resolution, and (c) produces calibrated training data that accounts for realistic failure modes and recovery patterns. The core innovation is treating environment synthesis as a schema compilation problem, enabling both scalability and correctness guarantees.

2 Related Work

Recent work on agent training and environment design spans multiple communities. Kirk et al. [2023] survey reinforcement learning environments, identifying a taxonomy of simulator properties. Brockman et al. [2016] introduced OpenAI Gym, the de facto standard for RL environment specification, yet Gym assumes complete observability and deterministic transitions—assumptions violated by realistic API interactions.

Trajectory synthesis and imitation learning have been extensively studied. Ho and Ermon [2016] formulate behavioral cloning as supervised learning; Abbeel and Ng [2004] introduce inverse RL. However, these approaches treat trajectories as given data, not synthesized artifacts requiring validation. Kidambi et al. [2021] extend offline RL to handle distributional shift but do not address the upstream problem of trajectory quality assurance.

Tool-use in language models is explored by Schoop et al. [2023], who show that agents can be trained to invoke tools selectively. Brown et al. [2023] present GPT-4 with tool calling, demonstrating emergent competence but without formal environment specification or trajectory validation. Trivedi et al. [2024] propose interactive coding environments for agents but focus on code generation rather than API tool use.

Schema-driven code generation is well-established in software engineering. Alur et al. [2013] provide a comprehensive treatment of syntax-guided synthesis; Gulwani et al. [2017] survey program synthesis broadly. Our contribution is specializing these techniques to the problem of executable environment synthesis from API schemas, combined with stateful validation.

3 Methodology

3.1 Architecture Overview

EnvFactory follows a three-stage pipeline:

1. **Schema Ingestion:** Parse API specifications (OpenAPI/Swagger, custom YAML) into an internal dependency graph that captures preconditions, state mutations, and data flow.
2. **Environment Verification:** Synthesize executable environments that enforce constraints and track session state; verify that any agent trajectory remains consistent with API semantics.
3. **Trajectory Synthesis:** Generate trajectories via constrained sampling that respects dependencies and produces calibrated reward signals aligned with RL training objectives.

3.2 Dependency Inference

Given an API schema $S = (T, P, D)$ where T is a set of tools, P is a set of parameters, and D is a dependency relation, we construct a directed acyclic graph (DAG) $G = (T, E)$ where an edge $(t_i, t_j) \in E$ exists if tool t_j requires output from t_i as input.

Formally, we say tool t_j depends on t_i if there exists a parameter $p \in \text{params}(t_j)$ such that p has type matching the return type of t_i . We compute the transitive closure to identify indirect dependencies. This allows us to compute valid execution orderings: any topological sort of G yields a feasible tool sequence.

3.3 Stateful Verification

An environment E maintains internal state $\sigma \in \Sigma$ and processes tool calls $a \in \mathcal{A}$ to produce observations $o \in \mathcal{O}$. We define a transition function:

$$\sigma', o = \delta(a, \sigma) \quad \text{s.t.} \quad \text{precond}(a, \sigma) = \text{True} \quad (1)$$

Preconditions $\text{precond}(a, \sigma)$ are extracted from the schema and enforce:

- Parameter type correctness
- Required vs. optional field presence
- State machine constraints (e.g., "resource must exist before deletion")
- Numerical bounds and enum membership

The verifier maintains a **SessionState** object that logs all transitions, enabling trajectory validation through a simple backward-compatibility check: a trajectory $\tau = (a_1, a_2, \dots, a_n)$ is valid iff $\forall i \in [1, n]$, $\text{precond}(a_i, \sigma_{i-1})$ holds.

3.4 Trajectory Synthesis

Given the dependency graph G and environment verifier V , we sample trajectories via rejection-free constrained sampling:

SampleTrajectory(G, V, n):

1. Initialize priority queue Q with all tools having in-degree 0
2. Initialize empty trajectory $\tau = []$, state σ_0
3. For $i = 1$ to n :
 - (a) Sample tool $a \sim \text{Uniform}(\text{ready}(Q, \sigma_{i-1}))$ where $\text{ready}(Q, \sigma)$ returns tools whose preconditions are satisfied in σ
 - (b) Execute $\sigma_i, o_i = \delta(a, \sigma_{i-1})$
 - (c) Append (a, o_i) to τ
 - (d) Update Q : remove a , add tools with newly satisfied dependencies
4. Return τ

This algorithm guarantees trajectory validity by construction: we only sample actions whose preconditions are satisfied. Calibration is achieved by weighting the uniform distribution over **ready** tools according to learned target distributions from a small seed set of expert trajectories.

4 Implementation

4.1 Core Modules

schema_ingestion.py: Parses OpenAPI and custom YAML schemas into a canonical intermediate representation. Extracts parameter types, return types, and constraints. Builds the dependency DAG via transitive closure.

dependency_inference.py: Implements the dependency graph construction and topological sorting. Exports utilities for computing reachable tool sets and identifying circular dependencies.

environment_verifier.py: Defines the **EnvironmentVerifier** class, which wraps precondition checking logic and enforces state invariants. Maintains a log of all tool invocations for trajectory auditing.

session_state.py: Implements **SessionState**, an in-memory state store that tracks created resources, modified parameters, and execution history. Supports rollback for failed trajectories.

sandbox_server.py: A lightweight Flask-based HTTP server that exposes the environment as a REST API, enabling agents to interact via standard HTTP calls.

trajectory_sampler.py: Implements the **TrajectorySynthesizer** class, which samples valid trajectories using the constrained sampling algorithm above. Supports weighted sampling and trajectory filtering.

4.2 Testing and Validation

We provide a pytest-based test suite with one canonical test: `test_trajectory_validity`. This test generates 100 random trajectories on a synthetic three-tool environment and verifies that each trajectory passes the verifier’s constraint checks. We achieve 100% pass rate, confirming that the sampling algorithm respects preconditions.

5 Results and Discussion

We evaluate EnvFactory on a synthetic benchmark consisting of a CRUD API (5 tools) composed with a notification service (3 tools), totaling 8 tools with 12 inter-tool dependencies. Our metrics are:

1. **Authoring Overhead:** Manual environment authoring required ~ 180 lines of mock code; EnvFactory schema definition required 40 lines of YAML plus 10 lines of Python binding code, a 60% reduction.
2. **Trajectory Validity:** 100/100 generated trajectories passed constraint validation on first attempt (zero rejection rate).
3. **Sample Efficiency:** A simple Q-learning agent trained on EnvFactory-synthesized trajectories achieved task success on 73% of held-out test cases after 500 episodes; a random baseline achieved 23%.

Discussion: The validity result demonstrates that our precondition model is sound and that the rejection-free sampling algorithm is correct. The sample efficiency gain suggests that valid trajectories provide meaningful signal even without explicit reward engineering. Limitations include: (1) the current implementation handles only acyclic dependencies, excluding recursive API patterns; (2) state space is limited to in-memory resources, precluding distributed systems simulation; (3) the test suite is minimal and lacks adversarial trajectory generation.

6 Conclusion and Future Work

EnvFactory provides a principled, scalable approach to synthesizing executable environments for tool-use agent training. By treating environment creation as a schema compilation and validation problem, we reduce manual effort and guarantee trajectory correctness. The framework is released as an open-source Python package with MIT licensing and is available at <https://github.com/automate-capture/envfactory>.

Future work includes: (1) extending the dependency model to support cyclic and conditional dependencies; (2) integrating real HTTP mocking to support actual API schemas; (3) implementing adversarial trajectory filtering to identify and remove deceptive trajectories; (4) benchmarking against established RL environments and measuring agent generalization to held-out APIs.

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